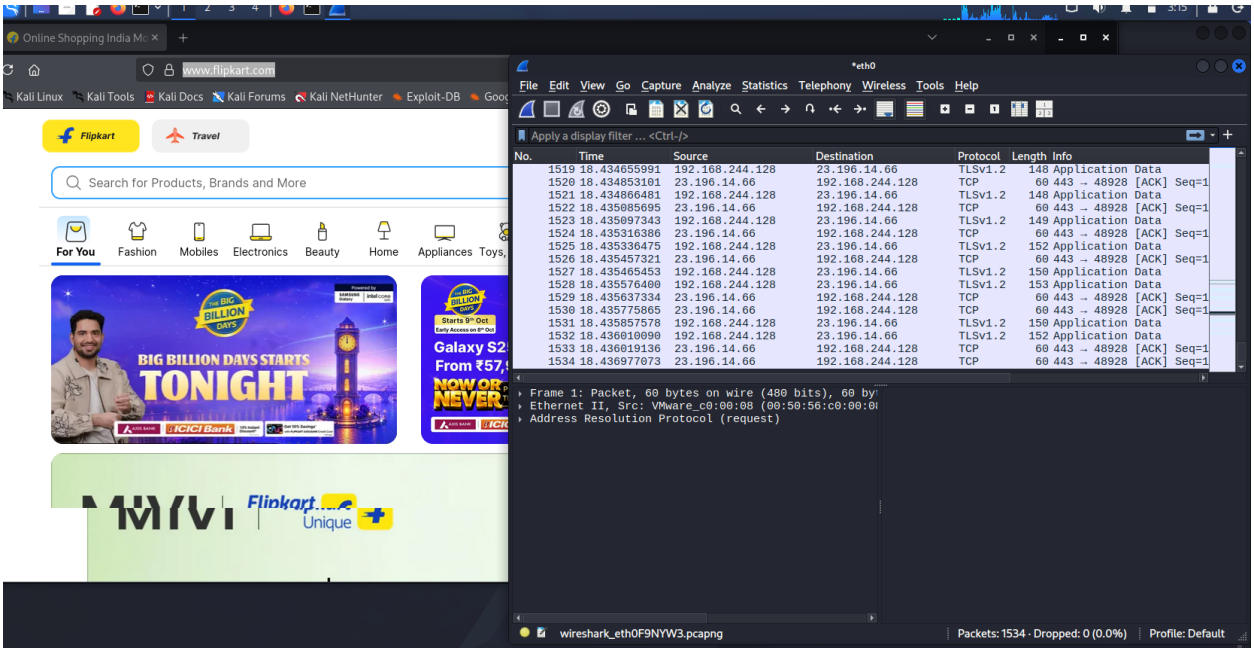
Assignment 13

TCP/IP Suite Theory

1. **Open Wireshark, capture network traffic while you refresh the Flipkart homepage, and identify one packet each examining belonging to the Application, Transport, Internet, and Network Access layers by their headers.**



Protocol names in packet list:

- **TLSv1.2 (Application Data packets)**
- **TCP (the [ACK] packets, 443 → 48928)**

1. Application Layer:

TLSv1.2 (Transport Layer Security)

2. Transport Layer:

TCP (Transmission Control Protocol)

3. Internet Layer:

IPv4 (Inferred from the Source/Destination IP addresses like 192.168.244.128)

4. Network Access Layer:

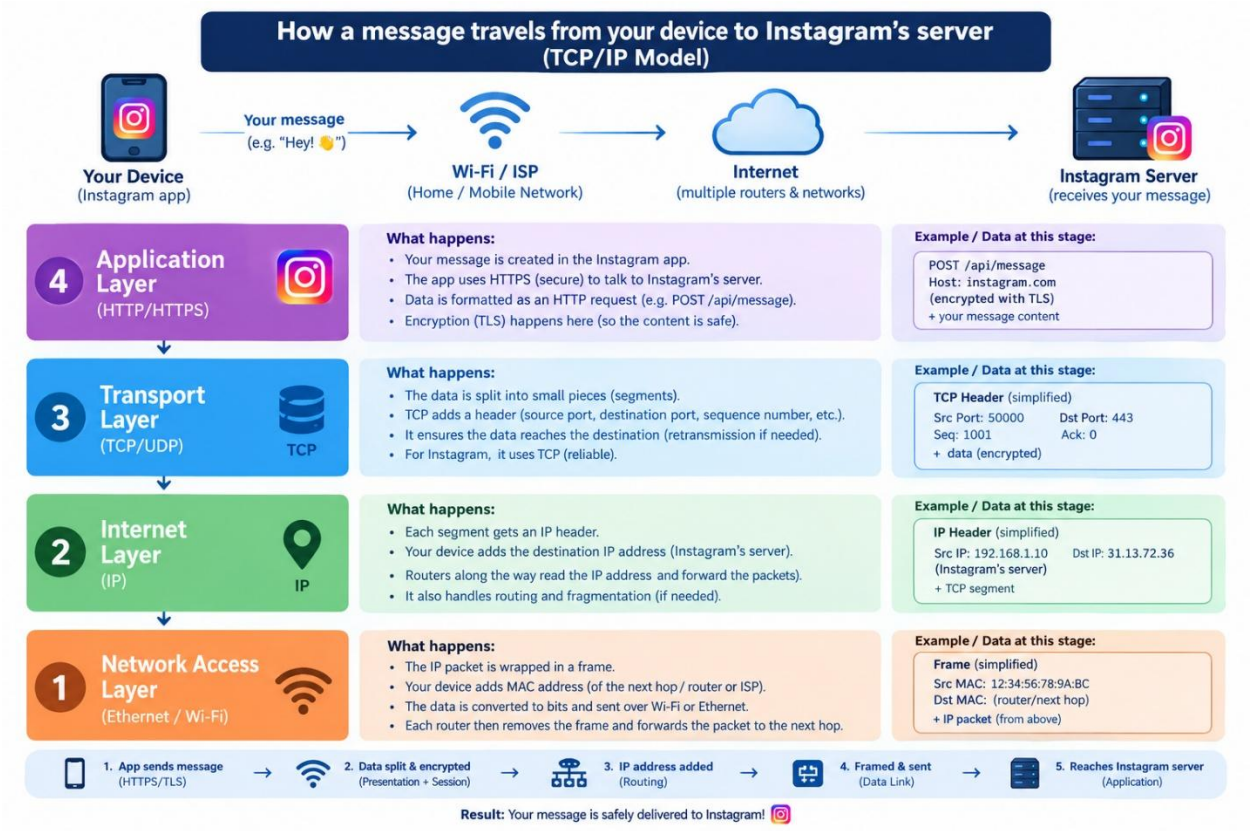
Ethernet II (Visible under the packet details pane at the bottom)

2. **Create a table comparing the OSI and TCP/IP models, listing all layer side-by-side, and for each layer, give a real-world example from any app you use (e.g., WhatsApp message, Zomato order).**

OSI Model	TCP/IP Model	Real-world example
7. Application	**Application**	WhatsApp: typing and sending a message
6. Presentation	**Application**	WhatsApp: message encryption/formatting
5. Session	**Application**	WhatsApp: maintaining the communication session
4. Transport	**Transport**	Message is broken into data pieces and delivered
3. Network	**Internet**	IP address helps the message reach the receiver
2. Data Link	**Network Access**	Wi-Fi/Ethernet sends frames between devices
1. Physical	**Network Access**	Wi-Fi radio signals or Ethernet electrical signals

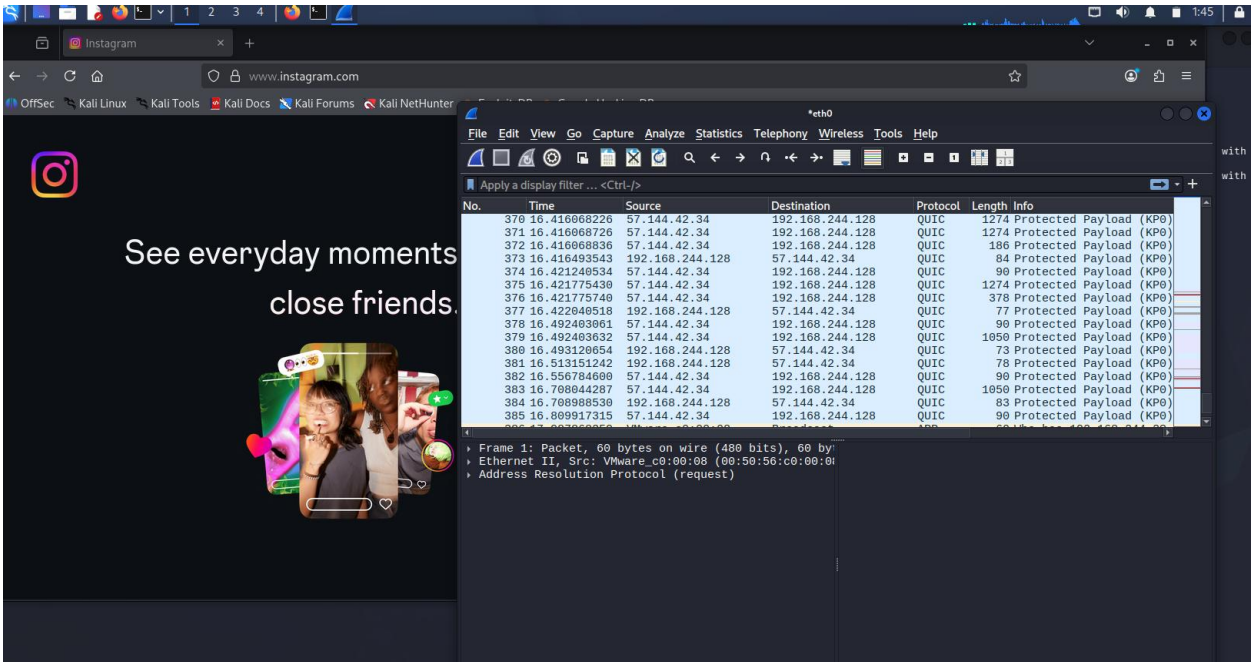
3. **Given a screenshot of a TCP segment header (provided by your trainer), label the fields that belong to the Transport layer and explain what each field does in 1-2 lines.**

4. **Draw a diagram showing how a message sent from your device to Instagram's server passes through each TCP/IP layer, and annotate what happens at each step using simple app-related terms (e.g., 'message split into packets', 'IP address added', etc.).**



5. **Pick any one packet you captured in Wireshark and write which TCP/IP layer it belongs to, how you identified it, and what real-world action in an app (like sending a WhatsApp message or booking a ticket on BookMyShow) it represents.**

❖ **Capture Instagram web-app traffic**



1. **TCP/IP Layer: Transport Layer (QUIC uses UDP)**

2. **How I identified it: I captured Instagram traffic in Wireshark and saw the protocol name QUIC in the Protocol column. QUIC works over UDP and provides fast, secure communication.**

3. **Real-world action: It represents Instagram app traffic, such as loading posts, reels, or other content from Instagram's servers. My capture shows QUIC traffic, but it does not prove that a specific message was sent.**